EDUCATIONAL DEMONSTRATION REFERENCE ? NOT AUTHENTIC INVESTIGATIVE RECORDS
Network forensics and suspicious connections

Sysmon Event ID 3 can record network connections and associate them with process identifiers/GUIDs when network connection logging is enabled. Event ID 22 records DNS queries when configured and supported. Review process identity, destination IP and port, protocol, initiating account, timing, DNS query/response context, and related process-creation events. Compare destinations and timing with expected application behavior and network baselines.

Potential beaconing leads include repeated connections at regular intervals, repeated small transfers, unusual destinations, and recurring DNS lookups. These patterns require context: legitimate health checks, proxies, NAT, CDNs, and scheduled tasks can look similar. Correlate endpoint logs with firewall, proxy, DNS resolver, packet capture, and server-side records when available.

Limitations: events may be disabled, filtered, lost, encrypted, or obscured by proxies, NAT, VPNs, shared hosting, or short retention. DNS caches and resolver logs have different coverage. A connection or domain lookup alone does not prove compromise, data transfer, actor identity, or intent.

Sources: Microsoft Sysmon documentation, https://learn.microsoft.com/sysinternals/downloads/sysmon; NIST SP 800-86, Guide to Integrating Forensic Techniques into Incident Response, https://csrc.nist.gov/pubs/sp/800/86/final.